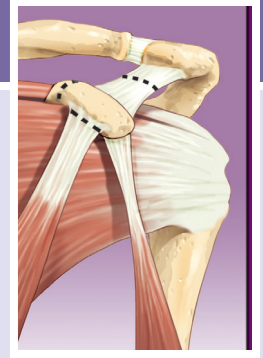


## Other Muscle Injuries

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Muscle injuries about the shoulder are increasingly common in active individuals, with the prevalence of upper extremity muscle sprains, strains, and ruptures rising as more athletes join gyms and engage in recreational and professional strength training and weight lifting.<sup>1-3</sup> However, these injuries remain inadequately described and defined despite their ability to cause significant functional disability, pain, and time lost from sport.<sup>1</sup>

### PECTORALIS MAJOR

Injuries to the pectoralis major muscle occur primarily in males 20 to 40 years of age, with 75% of cases being secondary to sporting activities.<sup>4</sup> Weight lifting has been shown to be responsible for approximately 50% of reported cases in the literature,<sup>5,6</sup> with injuries also being reported in athletes participating in wrestling, jujitsu, American football, and gymnastics.<sup>5-8</sup> Although the vast majority of patients are young males, injuries in females have been reported.<sup>9,10</sup> Injuries are typically sustained secondary to violent, eccentric muscle contractions,<sup>2</sup> with operative repair yielding significantly improved outcomes and restoration of preinjury strength and activity level as compared with nonoperative management.<sup>11</sup>

### Anatomy

The pectoralis major muscle covers the anterior chest wall and is composed of two separate heads (clavicular and sternal) that serve to adduct, internally rotate, and flex the shoulder.<sup>4</sup> The muscle originates as a broad sheet from the clavicle, sternum, ribs, and external oblique fascia, with the two heads converging and inserting over a 5-cm area on the lateral lip of the bicipital groove of the humerus.<sup>2</sup> The tendons are hypothesized to twist 90 degrees,<sup>11</sup> 180 degrees,<sup>12</sup> or simply to overlap at their insertion,<sup>13</sup> resulting in the inferior (sternal) segment inserting proximal to the more distal insertion of the superior (clavicular) segment on the humerus.<sup>2,4</sup> Because of this configuration, the fibers of the sternal segment experience increased excursion with the arm in 0 to 30 degrees of extension, as experienced during bench press, resulting in maximal stretching of the tendon fibers and increasing the risk for rupture.<sup>4,11,14</sup> The pectoralis major muscle is innervated by the medial (lower sternocostal segments) and lateral (clavicular segments) pectoral nerves arising from the C5 to T1 nerve roots.<sup>15</sup>

### Mechanisms

The majority of injuries to the pectoralis major are secondary to substantial contraction applied to the distal tendon while it is maximally stretched,<sup>2</sup> with direct trauma to the shoulder and arm being less common.<sup>6,11,16</sup> Multiple studies have found that the most common mechanism of injury was the bench press,<sup>3,17,18</sup> in which the abducted, extended, and externally rotated arm is eccentrically contracted, placing the pectoralis tendon under maximal tension.<sup>2,4</sup> Owing to the anatomic configuration of the tendons distally, application of maximal load results in a significant mechanical disadvantage to the fibers of the inferior sternal segment of the pectoralis major, thus often causing them to rupture and accounting for the high incidence of sternal head ruptures.<sup>11,17</sup> Although partial ruptures are more common, up to 65% of complete ruptures have been reported to occur at the humeral insertion, with up to 29% occurring at the musculotendinous junction.<sup>5,6</sup> Moreover, distal tendon injuries are more likely to be complete ruptures.<sup>2</sup> Injuries to the muscle belly or the clavicular and sternal origins of the pectoralis major are less common and usually result from direct trauma.<sup>1</sup>

Other mechanisms—including direct trauma sustained during American football and rugby<sup>3</sup> along with other sporting activities associated with a risk of rapid, forceful shoulder abduction, such as rodeo, windsurfing, sailing, handball, wrestling, hockey, and artistic gymnastics<sup>19-23</sup>—have been described. In addition, injuries have also been described in individuals attempting to break a fall in which the force of impaction on the ground on the contracted muscles results in injury.<sup>1,2</sup>

Abuse of anabolic steroids has been tied to an increased risk for tendon ruptures.<sup>24,25</sup> Pochini et al.<sup>3</sup> have reported that in 60 patients with complete ruptures, more than 90% endorsed anabolic steroid use, whereas Aarimaa et al.<sup>19</sup> point to a 36% rate of abuse in their case series of 33 patients. Other authors have proposed that steroids weaken and stiffen the tendon fibers secondary to tendinopathy.<sup>3,19,24</sup> Concomitant with disproportionate gains in muscle mass and force,<sup>3,6,8,9,19</sup> the weakened tendon is overwhelmed, resulting in rupture with continued stress placed across the tendon with weight lifting.

### Classification

Injuries to the pectoralis major are generally classified using the classification system proposed by Tietjen et al.,<sup>26</sup> which describes

## Abstract

Muscle injuries about the shoulder are increasingly common among active individuals, with the prevalence of upper extremity muscle sprains, strains, and ruptures rising as more athletes join gyms and engage in recreational and professional strength training and weight lifting. However, these injuries remain inadequately described and defined despite their ability to cause significant functional disability, pain, and time lost from sport.

## Keywords

athlete  
baseball  
latissimus dorsi  
pectoralis major  
rupture  
sport  
steroids

the site of the lesion without accounting for the severity (partial versus complete), timing (acute versus chronic), or segments of involvement (sternal versus clavicular head).<sup>3,4</sup> A recent meta-analysis by El Maraghy and Devereaux<sup>5</sup> proposed a new classification scale focusing on lesion acuity (acute or chronic), the qualitative degree of tendon involvement (incomplete versus complete), and the location of the rupture (muscle, musculotendinous junction, or humerus).

## Physical Examination

In addition to a thorough history, a comprehensive physical examination is essential for diagnosis and to ensure proper clinical decision-making while ruling out concomitant injuries in and around the shoulder. Acute ruptures to the distal tendons present with edema, ecchymosis, and hematoma to the lateral chest and proximal arm, while injuries to the sternal or clavicular origins are isolated over the anterior chest wall. Patient typically report hearing or feeling a popping or tearing sensation at the time of injury, accompanied by pain in the chest wall, axilla, or down the arm.<sup>4,27</sup> Examination will demonstrate weakness with concomitant discomfort with resisted adduction, forward flexion, and internal rotation, along with tenderness at the axilla and humeral insertion.<sup>1,4,28</sup> Further inspection may reveal an asymmetric prominence from hematoma or from retraction of the muscle belly near the axilla.<sup>4</sup> As swelling subsides, examination of the axillary fold may demonstrate a palpable defect in the anterior axilla, accompanied by thinning, hollowing, or loss of the fold accentuated with isometric contraction with adduction as the patient presses the hands against one another in the front of the chest.<sup>8,11</sup> Patients may also demonstrate a visibly retracted stump within the anterior axillary fold on lateral inspection of the arm during forward flexion, known as the “S” sign.<sup>4</sup> Asymmetric webbing of the axilla is commonly reported to describe the loss of musculature from the anterior wall of the axilla.<sup>10</sup> Whereas some authors describe the ability to differentiate full-thickness tears from partial tears by the presence of a distinct gap or defect,<sup>4</sup> lack of a palpable defect in the axilla has not been shown to be a reliable indicator of pectoralis continuity.<sup>1</sup> In cases of chronic tendon injuries, patients may develop a prominent skin fold, known as *cicatricial fibrosis*, along with webbing and prominence of the distal deltoid insertion.<sup>7,11</sup> Retraction of the tendon and muscle with appreciable asymmetry is more pronounced as compared with the intact contralateral side in chronic tears as well.

## Imaging

In the setting of suspected injury to the pectoralis major, plain radiographs are of limited use; however, they are essential to assess for bony abnormalities and avulsions, which are reported to occur in 2% to 5% of cases.<sup>5,6,10</sup> Furthermore, radiographs allow for assessment of the glenohumeral joint, as concomitant dislocations of the shoulder have been reported.<sup>29</sup> Magnetic resonance imaging (MRI) is the modality of choice for assessing the site (muscle belly, tendon, or myotendinous junction) and degree of tendon injury,<sup>6,12,25,30</sup> with axial oblique and coronal views offering the best visualization.<sup>2</sup> Acute tears can be distinguished from chronic tears on MRI by high signal intensity at

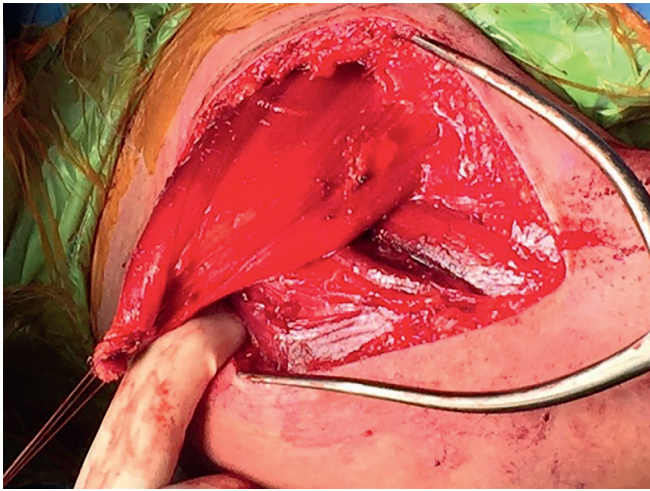
the musculotendinous junction on T2-weighted images owing to hemorrhage and edema with visualization of tendon and muscle retraction or tendon-bone discontinuity.<sup>25,30</sup> Chronic tears demonstrate low signal intensity and muscle retraction, suggestive of fibrosis and scarring.<sup>30</sup> Studies reporting on the correlation of findings on MRI with those of surgery have found excellent correlations based on the extent and location of injury.<sup>2,12,31</sup> MRI of the chest wall instead of the shoulder along with external rotation of the arm may improve visualization and aid in proper diagnosis.<sup>4,32</sup> However, investigations have found that MRI may not successfully identify or allow for complete differentiation of the affected portions of the pectoralis muscle or tendon.<sup>12,25</sup> Therefore advanced imaging modalities should serve only to complement the clinical examination.

## Treatment

The treatment of pectoralis injuries depends on the extent and location of tearing as well as the activity level of the patient. Conservative management is recommended for strains, contusions, muscle belly ruptures, and partial tears as well as complete tears in low-demand sedentary individuals.<sup>4,6</sup> Treatment involves rest, ice, pain control, and sling immobilization with the arm adducted and internally rotated.<sup>8,33</sup> Patients treated conservatively have demonstrated good functional outcomes with the ability to perform activities of daily living and no significant limitations.<sup>11,17,34</sup> Reported complications include hematoma, abscess formation, incision breakdown, and myositis ossificans as well as cosmetic defects or persistent deficits in strength, especially in younger patients.<sup>33,35</sup>

Surgery is reserved for young, active individuals suffering complete tears, including those to the sternal head/posterior lamina,<sup>9</sup> tears at the myotendinous junction, intratendinous tears, or those at the tendon insertion.<sup>4</sup> Surgery is recommended to allow for restoration of normal strength, allowing active athletes to successfully return to sport at preinjury levels.<sup>2,6,20</sup> Multiple techniques for fixation in the acute period have been reported, including suturing tendon to periosteum<sup>17</sup> or remaining tendon<sup>16</sup> in addition to bony fixation using drill holes through the humerus fixed with anchors,<sup>36</sup> screws, and washers<sup>7</sup> or endosteal buttons (Fig. 50.1). Studies have found that young, active patients treated surgically experience significantly better excellent/good functional outcome scores,<sup>2,6,7</sup> improvement in strength,<sup>11,37</sup> and patient satisfaction,<sup>23</sup> cosmesis, pain control, and return to sport as compared with those treated nonoperatively.<sup>3,6,7,11,27,33,37</sup> Reported complications following operative management include post-operative stiffness, hypertrophic scarring, infection, and recurrent tearing.<sup>2,11,18,38</sup>

The timing from injury to repair has been shown to influence the ease of repair and subsequent outcomes, with authors recommending surgical treatment as soon as possible.<sup>3</sup> Currently, no consensus exists on the definition in timing for acute tears, with multiple studies citing 2 weeks,<sup>23</sup> 3 weeks, 6 weeks,<sup>19,39,40</sup> and up to 8 weeks.<sup>6</sup> Compared with outcomes following operative management of chronic tears, authors have cited statistically better outcomes with treatment occurring soon after injury in the acute phase.<sup>3,6,19,39</sup> This is because the repair of chronic injuries requires increased surgical exposure and dissection owing to



**Fig. 50.1** Intraoperative photo demonstrating fixation of acute pectoralis major tearing using suture fixation to the distal tendon.

muscle retraction and fibrosis.<sup>5</sup> Moreover, direct repair can be difficult or impossible, necessitating reconstruction using allograft or autograft augmentation.<sup>4</sup> Reconstruction has been described using human- and animal-derived patches, hamstring augmentation,<sup>7</sup> Achilles tendon allograft,<sup>41</sup> fascia lata allograft,<sup>42</sup> flexor tendon grafts,<sup>43</sup> and bone patellar tendon autograft.<sup>44</sup>

## LATISSIMUS DORSI

Injuries to the latissimus dorsi are rarely reported; they are limited to case reports, small case series, and retrospective analyses. Ruptures have been noted to occur in baseball pitchers, golfers, and participants in rodeos, tennis matches, wrestling, water skiing, and rock climbing.<sup>45–53</sup> Complete tendon ruptures are generally season-ending injuries with lengthy rehabilitation and the potential for significant disability.<sup>53</sup>

### Anatomy

The latissimus dorsi is a large triangular muscle originating from the spinous processes and supraspinous ligaments of T7–T12 along with the lumbar and sacral vertebrae, lumbar fascia, posterior third of the iliac crest, four most caudal ribs, and inferior angle of the scapula. The muscle spans across the posterior thoracolumbar region, converging into a thick fasciculus transitioning into a tendon that inserts on the floor of the bicipital (intertubercular) groove of the humerus via a conjoint tendon with the teres major.<sup>54</sup> The latissimus dorsi functions to adduct, extend, and internally rotate the humerus in addition to lifting the trunk forward and upward during climbing. Innervation is provided by the thoracodorsal nerve coming off of the posterior cord of the brachial plexus, which runs with the thoracodorsal artery, innervating the muscle on its anteroinferior surface approximately 2 cm medial to its muscular border.<sup>55</sup> Pearle et al. reported that the thoracodorsal pedicle, consisting of the artery and nerve, inserted on the anterior part of the latissimus dorsi muscle belly an average of 13.1 cm (range, 11.0 to 15.3 cm) medial to its humeral insertion.<sup>56</sup>

### Mechanisms

Strains, tears, and ruptures to the latissimus are generally sustained from a sudden power extension or hyperabduction force applied to the arm or shoulder.<sup>49</sup> The majority of reported cases have occurred in baseball pitchers,<sup>45,53</sup> in whom the latissimus dorsi has been shown to be active during several phases of the pitching cycle, especially during late cocking and serving. These movements are among the most powerful generators of force during the acceleration phase.<sup>57</sup> The role of the latissimus during pitching effectively places an eccentric force across the distal tendon attachment, increasing the risk of injury and rupture.<sup>53</sup> Moreover, a number of pitchers report sensing the injury and most of the pain during the follow-through phase, accounting for the continued role of the latissimus during the deceleration phase of pitching.<sup>53</sup> Activation of the latissimus during the pitching cycle has been noted to be higher among professional pitchers versus amateur pitchers.<sup>58</sup> Athletes generally report a “popping” or tearing sensation to the posterior aspect of the shoulder; however, evolving tightness and fatigue to the posterior shoulder have also been reported.<sup>53</sup>

### Classification

Because of the rarity of the injury, no classification system in the literature has been proposed to date.

### Physical Examination

Athletes generally seek medical attention as a result of loss of power with pain in the shoulder that limits athletic and overhead activities. Athletes may also complain of a burning pain to the superior aspect of the triceps area, along with ecchymosis.<sup>52</sup> The most significant deficit in strength has been reported in shoulder extension.<sup>48</sup> In addition, individuals will possess decreased strength with both isometric internal rotation and adduction of the arm; however, athletes may maintain excellent strength with these movements secondary to the recruitment of adjacent muscles.<sup>52</sup> A palpable mass, indicative of a retracted latissimus muscle belly or hematoma, may be appreciated to the posterolateral aspect of the chest and shoulder girdle<sup>52</sup> (Fig. 50.2A and B), while the asymmetry is more pronounced with the arms abducted on the hips or with adduction and internal rotation of the shoulder against resistance.<sup>49,52</sup> Such individuals are generally without any neurologic deficits.

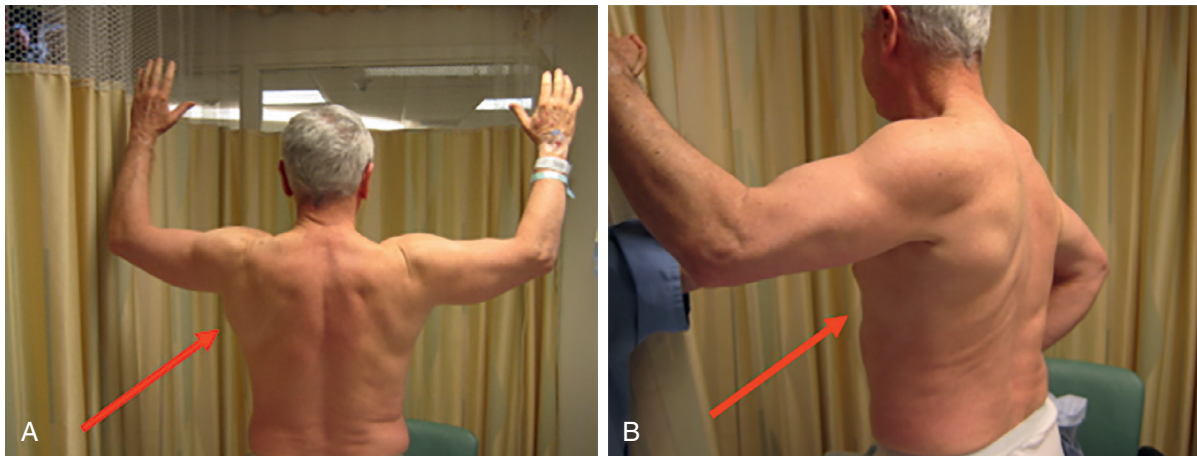
### Imaging

No investigation has reported on the benefit of plain radiographs in the diagnosis of latissimus dorsi ruptures; however, ruling out fracture and dislocation is critical. Multiple studies have reported MRI as the most reliable modality to confirm diagnosis while also demonstrating the extent of injury along with ruling out concomitant lesions to the shoulder girdle (Fig. 50.3A–C).<sup>49,53,59,60</sup> Ultrasound in trained hands has also been reported to enable detection of muscle lesions.<sup>59,60</sup>

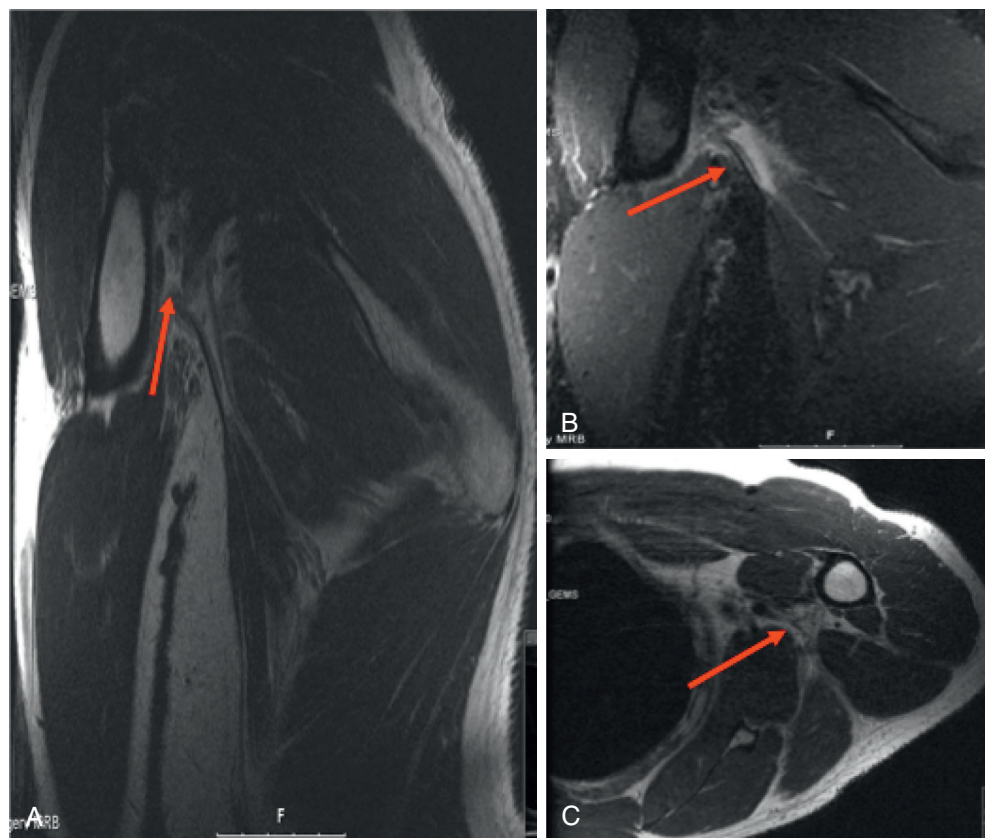
### Treatment

Successful management of latissimus dorsi injuries has been reported with both operative<sup>49,54,61</sup> and nonoperative treatment.<sup>46,52,53</sup>





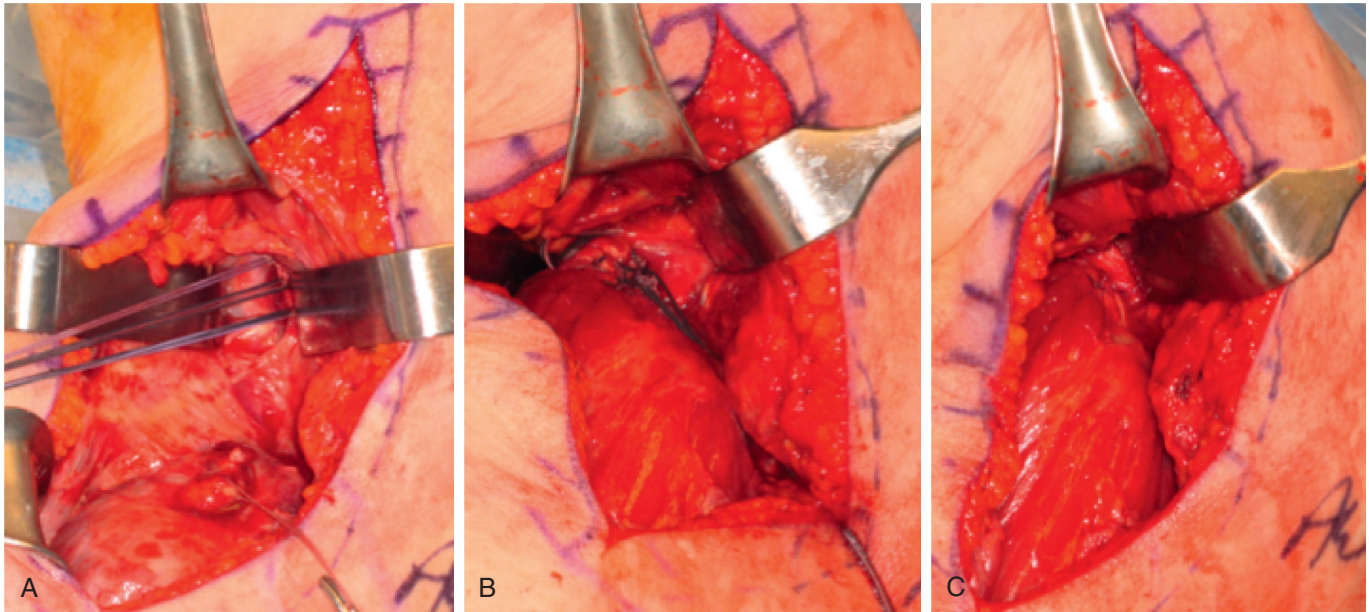
**Fig. 50.2** Physical examination findings showing a palpable mass (*red arrows*) along the posterolateral aspect of the shoulder girdle as seen from the posterior and lateral views, indicative of a retracted latissimus dorsi rupture.



**Fig. 50.3** Magnetic resonance imaging demonstrating complete rupture of the latissimus dorsi tendon (*red arrows*) in the sagittal (A), coronal (B), and axial (C) planes.

However, owing to the infrequency of reported cases, there is no consensus on the standard of care for these injuries or whether surgical intervention provides any advantages as compared with conservative management. In their treatment of 10 professional baseball pitchers including 3 with tendon avulsions and 7 with strains, Nagda et al.<sup>53</sup> reported that all but one athlete returned to the same or a higher level of play following rest for up to 28 days in addition to rehabilitation with advancement to a supervised throwing program. Athletes sustaining strains were reported to

return to play (RTP) after 12 weeks and those with avulsions after 16 weeks. The authors did note, however, that avulsions were season-ending injuries and that the three pitchers who began throwing soonest experienced setbacks that prolonged the delay in their return to sport. Advocates of surgical intervention cite the restoration of preinjury strength and function in high-level athletes<sup>48,51,62</sup> with repairs becoming technically more challenging 2 or months following injury.<sup>63</sup> Operative techniques included repair of the tendon to the humeral footprint utilizing



**Fig. 50.4** Intraoperative photo demonstrating (A) placement of suture within the torn distal aspect of the latissimus dorsi followed by (B) placement of free suture ends through suture anchors to recreate the spiral anatomy of the tendon insertion with (C) final repair.

transosseous tunnels<sup>47,51,63</sup> or suture anchors (Fig. 50.4A–C).<sup>62,63</sup> In the limited studies reported in the literature, operative management has demonstrated good functional outcomes and restoration of strength in athletes,<sup>46,49</sup> with RTP usually reported 5 to 6 months following repair.

### Teres Major

The teres major functions in pulling objects toward the body and lifting the body while climbing. The small muscle originates from the dorsal surface off the inferior angle of the scapula and inserts on the bicipital groove of the humerus via a conjoint tendon with the latissimus.<sup>54</sup> The largest series of cases, by Nagda et al.,<sup>53</sup> comprised 11 pitchers, among whom isolated injuries to the teres major were found in 6 (three avulsions and three strains), a combined latissimus/teres major avulsion in 1, and combined latissimus dorsi/teres major strains in 4. All were treated nonoperatively, with only one athlete not returning to play following arthroscopic superior labrum anterior and posterior (SLAP) repair. Other reported injuries to the teres major in the athletic population include an avulsion in a baseball pitcher<sup>64,65</sup> and another in a water skier who sustained an isolated tear caused by the arm being forced forward from a tow rope.<sup>66</sup> Although the teres major has been speculated to serve a function similar to that of the latissimus dorsi during the pitching cycle, no study examining the exact contribution of the teres major has been

performed.<sup>53</sup> MRI is used to differentiate between injuries to the teres major versus latissimus dorsi, as individual discrimination on physical examination is difficult.<sup>65</sup> Although no consensus regarding optimal treatment has been presented in the literature, all reported cases in baseball pitchers have been treated nonoperatively with rest, rehabilitation, and gradual return to sport without any reported complications.

For a complete list of references, go to [ExpertConsult.com](http://ExpertConsult.com).

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